

SMPS Rev-A Design Session — Extensive Technical Notes

1. Project Overview

Offline isolated flyback SMPS converting 120 VAC to isolated 5 V output.

Controller: Power Integrations TNY285.

Isolation via custom hand-wound transformer.

Regulation via TL431 + optocoupler.

Design intent: professional Rev-A board suitable for fabrication, bring-up, and portfolio use.

2. Transformer Handling (T1)

Transformer is custom, hand-wound.

Lifecycle set to Not Managed.

No MPN or supplier assigned.

This is industry-correct handling for custom magnetics and does not block fabrication.

3. BOM & ActiveBOM Interpretation

ActiveBOM warnings such as 'No Solutions' and 'No MPN Ranked' indicate missing supplier metadata only.

They do NOT indicate electrical errors and do not block PCB fabrication or assembly.

4. Schematic Architecture

Input stage includes mains EMI suppression and bridge rectifier.

Post-bridge HV domain clearly separated from secondary.

Drain node identified as highest dv/dt region.

Aux winding provides controller bias.

Secondary includes LC filtering, USB output, and ESD protection.

5. Net Naming Cleanup

Auto-generated NetC*/NetD*/NetR* nets were replaced with functional names.

This enabled accurate net class assignment and rule definition.

This step significantly improved design clarity and professionalism.

6. Final Net Classes

MAINS_PRIMARY: mains, bulk HV, primary return, EMI choke, aux bias.

DRAIN_HOT: Drain node only.

SECONDARY_POWER: 5 V output and secondary ground.

SIGNALS_IO: control, feedback, USB data, injection points.

7. Layout Rule Strategy

Isolation clearance enforced between MAINS_PRIMARY and SECONDARY_POWER.

Drain node given extra clearance and no copper pour.

Secondary ground allowed solid pours.
Signal nets routed thin and away from hot nodes.

8. Fabrication Readiness

Custom transformer footprint verified.
Creepage slots present.
Net classes defined.
Design is ready for Gerber export and PCB fabrication.

9. Professional Assessment

Design reflects industry-standard offline SMPS practices.
Clear domain separation.
Appropriate handling of custom components.
Strong portfolio-quality hardware project.

10. Recommended Next Steps

Run final DRC.
Export fabrication package.
Prepare bring-up checklist.
Archive Rev-A as frozen baseline.